AttnMove: History Enhanced Trajectory Recovery via Attentional Network

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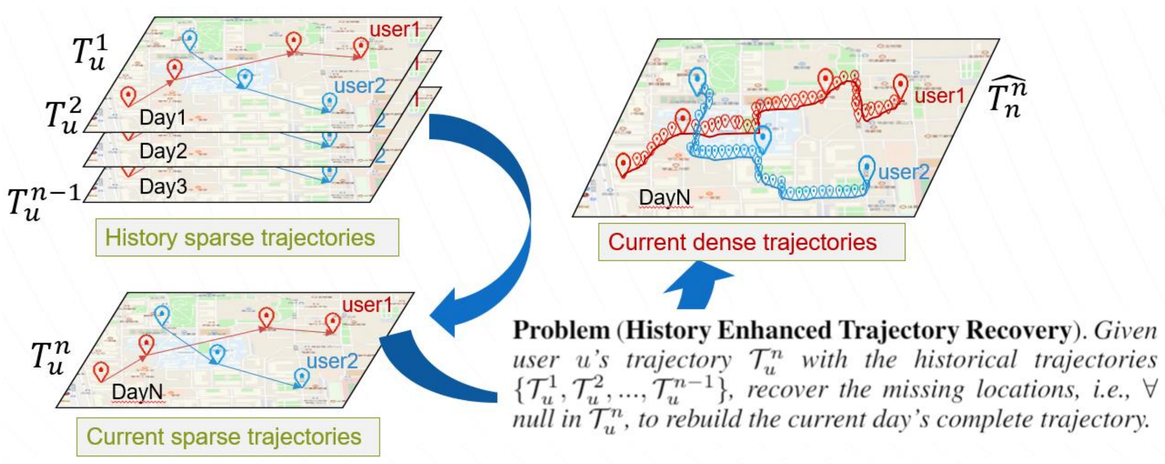
Introduction

Research question



Problem (History Enhanced Trajectory Recovery). *Given user u's trajectory $\mathcal{T}_u^n$ with the historical trajectories $\{\mathcal{T}_u^1, \mathcal{T}_u^2, ..., \mathcal{T}_u^{n-1}\}$, recover the missing locations, i.e., $\forall$ null in $\mathcal{T}_u^n$, to rebuild the current day's complete trajectory.*

Three ***challenges*** for History enhanced trajectory reconstruction :

- High sparsity results in weak spatial-temporal constrains
- Periodical feature is hard to distill from noisy history trajectories
- It intractable to combine historical pattern with observed locations

Method

System Framework

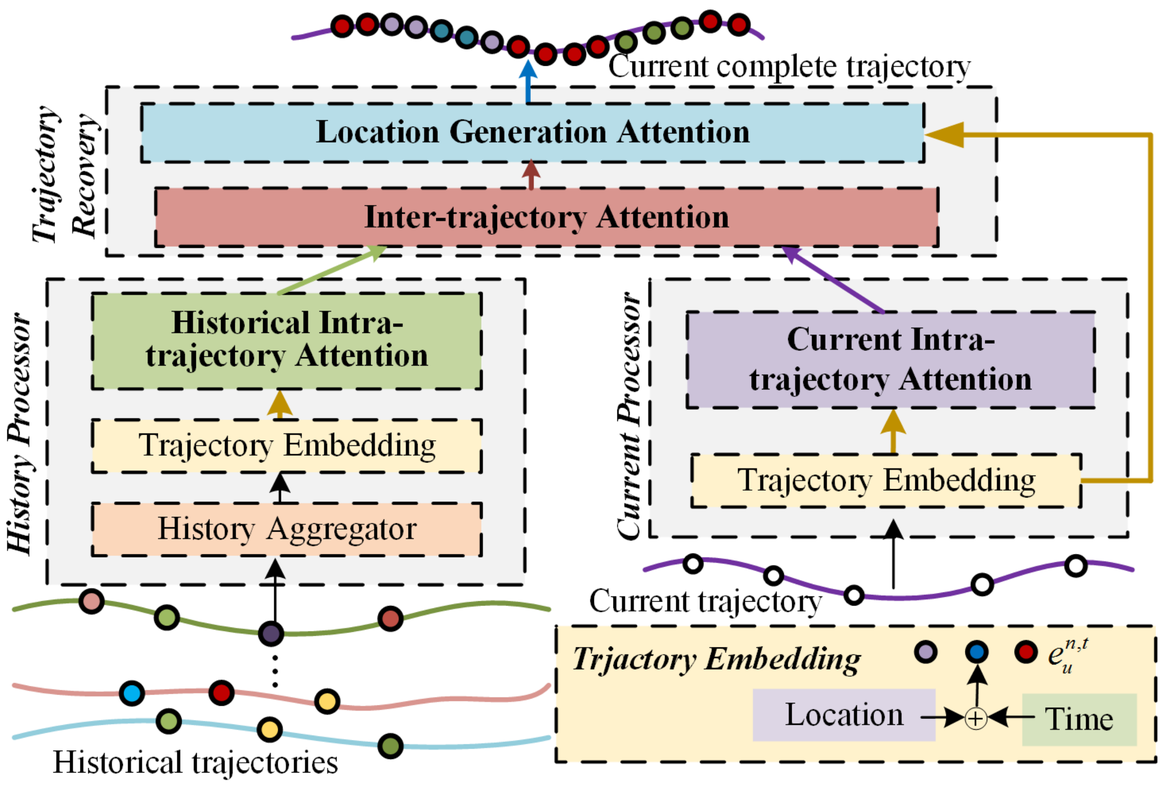
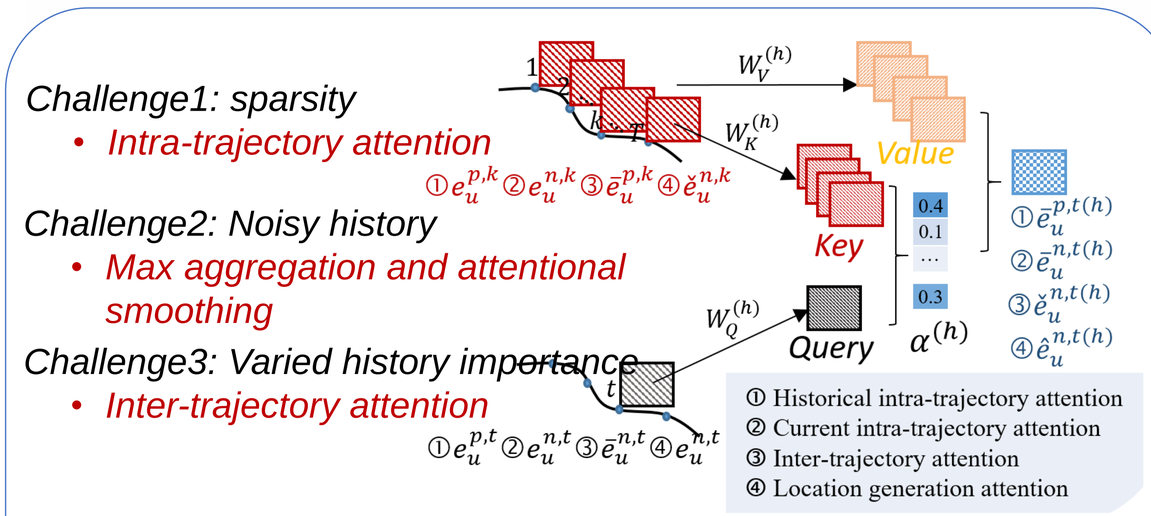


Figure 1: Main architecture of the proposed AttnMove

Modules:

- *1.Mobility representation: dense representation for time and location*
- *2. Current constrain learning: enhance spatial-temporal constrains*
- *3. Historical pattern learning: extract historical pattern*
- *4. Fusing and reconstruction: combine history and current trajectory*

Code and data: https://github.com/XTxiatong/AttnMove



Experiments

Dataset

Table 1: Basic statistics of mobility datasets.

Dataset	City	Duration	Users	Trajectories	Regions	Regions/day
Tencent	Beijing	1 month	4265	39,422	8,998	6.4
Geolife	Beijing	5 years	40	896	3,439	9.1

➢ Evaluation methodology

- Map GPS coordinates into urban regions, i.e., blocks demarcated by road networks, and discrete time into 30 minutes.
- Randomly mask 30 and 10 time intervals of each, the overall remained number is less than 10 (i.e., 20% of one day).
- Sparse trajectories as the input, and masked locations as ground truth
- Sort each user's trajectory by time, 7:1:2 for train, validation and test

Results

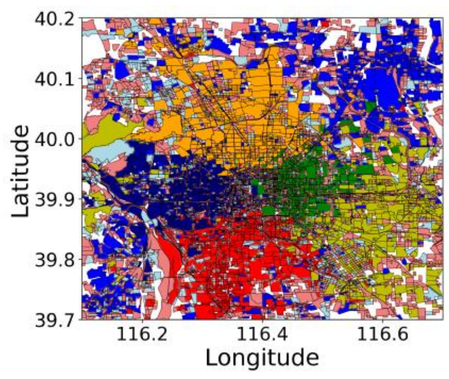
Table 2: Overall performance comparison. The best result in each column is in bold, while the second is underlined.

Model	Tencent			Geolife		
	Recall	*MAP*	*Dis.(m)*	*Recall*	*MAP*	*Dis.(m)*
TOP	0.5879	0.6123	2530	0.2757	0.2879	5334
History	0.4724	0.4937	1613	0.2505	0.2648	5116
Linear	0.6234	0.6567	1145	0.3642	0.3889	**2383**
RF	0.4848	0.4912	8128	0.2551	0.2540	6144
LSTM	0.6084	0.6722	3759	0.2725	0.3314	5864
BiLSTM	0.7090	0.7805	1371	0.3471	0.4148	5097
DeepMove	0.7259	0.7872	1322	0.3391	0.4015	4912
Bi-STDDP	0.7037	0.7831	1168	0.3701	0.4510	4041
AttnMove-H	0.7358	0.7999	1083	0.3853	0.4501	4129
AttnMove	**0.7646**	**0.8249**	**934**	**0.3982**	**0.4691**	3886

✓ *4.0%~7.6% performance gain compared to SOTA*

✓ *Superiority of Attentional Network*

✓ *Better utilization of history*



(a) Location embedding

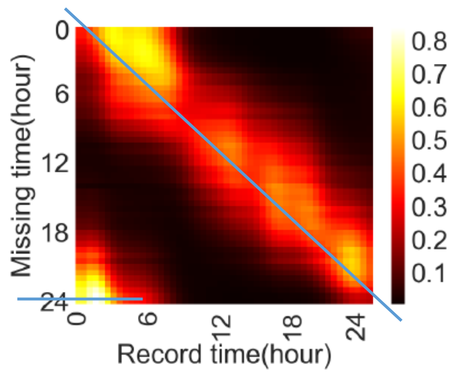


Figure 8: Attention weight

LAB INFO
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